

Cooperative Communication via Automated Guided Vehicle and Unmanned Aerial Vehicle: A Distributed Collaborate Beamforming Method

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Abstract—The advantages of autonomy, stability, and high load capability make automated guided vehicles (AGVs) appealing for applications like intelligent transportation networks. Nevertheless, AGVs may face limitations in terms of flexibility and transmission efficiency. In this paper, we consider the scenario where unmanned aerial vehicles (UAVs) serve dual roles in enhancing the communications of AGVs. Specifically, one group of UAVs is employed to support AGVs in data transmission, while another group of UAVs equipped with computational resources, functions as aerial base

stations (ABSs) for receiving and processing the collected data. Following this, we explore the collaborative deployment between AGVs and UAVs, and propose a highly efficient, low-interference, and energy-efficient uplink data transmission framework based on distributed collaborative beamforming. Correspondingly, we formulate a high-performance and low-interference transmission multi-objective optimization problem (HLMOP) to reduce the transmission time and operation energy consumption of the AGVs and UAVs, while minimizing the total sidelobe levels toward the directions of all non-current receiving ABSs. Due to the NP-hardness of the HLMOP, we propose a swarm intelligence algorithm, namely, improved multi-objective ant lion optimization (IMOALO), with three improved operators. Simulation results show that the proposed IMOALO algorithm performs better and can generate more excellent solutions than other benchmark algorithms.

Index Terms—Automated guided vehicle, unmanned aerial vehicle, distributed collaborative beamforming, multi-objective optimization.

I. INTRODUCTION

WITH the rapid integration of wireless communications in traditional fields, various industries have ushered in an era marked by digital, intelligent, and automated advancements. In intelligent transportation, automated guided vehicle (AGV) networks have attracted widespread attention [2]. Specifically, the main purpose of AGV networks is to improve communication and collaboration between vehicles to enhance traffic safety, improve traffic efficiency, and provide new traffic services. For instance, AGVs can share vehicle-related information, such as location and speed, to control the signal system. By analyzing this data, the intelligent transportation system can adjust intersection traffic signals according to real-time traffic flow, thereby effectively reducing waiting times and maximizing road capacity [3]. Moreover, some potential hazards on the road can be detected by coordinating communications and sensors between AGVs and roadside infrastructures, thereby effectively mitigating the probability of traffic accidents [4]. Despite the promising applications, the development of the AGV network still faces some critical issues, such as the weak communication capability of AGVs and limited coverage of roadside infrastructures, which limit the flexibility and application range of such systems in complex scenarios [5].

Unmanned aerial vehicles (UAVs) can be regarded as a novel type of intelligent transportation infrastructure that can support

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terrestrial traffic networks [6]. Specifically, UAVs offer the advantages of high mobility and flexible deployment, thus they can be deployed to assist AGVs in reducing communication costs [7]. Additionally, UAVs are able to serve as aerial base stations (ABSs), playing a pivotal role in collecting information such as traffic flow, vehicle speed, and road conditions [8]. Therefore, UAVs have the potential to form an AGV-UAV network, thereby helping AGVs perform data transmission or forwarding tasks to ABSs. However, the antenna performance of both AGVs and UAVs is impacted by their payload capacity and installation space, limiting the transmit power of a single AGV or UAV. Moreover, intelligent transportation systems have a multitude of communication devices, and communication in AGV-UAV networks may cause significant interference to these devices. Consequently, it is necessary to investigate a novel method to achieve high-performance and low-interference transmission between the AGV-UAV system and remote ABS.

Distributed collaborative beamforming (DCB) efficiently enhances the transmission performance in distributed multi-antenna systems [8], [9]. Specifically, the fundamental principle of DCB is to emulate classical beamforming and harness the cooperative capabilities of multiple independent systems, such as AGVs or UAVs, to enhance the communication abilities. For instance, we utilize the antennas of multiple AGVs and UAVs to compose a VAA, and then their electromagnetic waves create constructive and countervailing interference in different directions. Consequently, the VAA can generate a beam pattern with a high-gain mainlobe and several low-signal-intensity side lobes. Compared with the classical beamforming which is based on a fixed antenna array, the primary advantage of such VAA is its ability to exploit the maneuverability and flexibility of these vehicles. For instance, the number of AGVs and UAVs in the VAA can be dynamically adjusted according to the transmission distance, thereby facilitating high-quality communication at reduced costs. As such, by carefully designing the beam pattern, the VAA can point the mainlobe toward the ABSs and suppress interference with other devices. Therefore, constructing the AGVs and UAVs as VAA shows promise in improving transmission performance while reducing interference.

However, optimizing a qualified beam pattern for such systems is not a straightforward task. Specifically, the excitation current weights and positions of the AGVs and UAVs within the VAA are key factors that impact DCB performance and should be carefully optimized. To this end, AGVs and UAVs need to move to better positions for performing DCB. However, this operation process generates more energy, thus shortening the lifetime of AGV-UAV data transmission systems. Such trade-offs between energy efficiency and transmission performance need to be analyzed. Moreover, in intelligent systems, AGVs need to transmit critical data to multiple ABSs for data security and backups. Thus, scheduling the transmission process to further enhance energy efficiency needs to be considered.

Accordingly, our aim is to conduct a multi-objective optimization framework to balance the transmission performance and energy efficiency by optimizing the numerous decision variables that are related to AGVs and UAVs, as well as the system scheduling. The key contributions of this paper are outlined as follows:

- *Joint AGV-UAV DCB Data Transmission System:* We enhance the performance of traditional AGV data transmission systems by integrating UAVs and DCB. Specifically, we form the AGVs and assisted UAVs as a VAA to enhance the transmission gain to ABSs and mitigate interference towards other devices. Such systems are particularly useful in intelligent transportation networks, remote area communications, and high-mobility environments. To the best of our knowledge, this is the first work to conduct joint DCB optimization of AGVs and UAVs.
- *Mixed-Variable Multi-Objective Optimization Problem:* We design a high-performance and low-interference transmission multi-objective optimization problem (HLMTOP) aiming to minimize the total transmission time, total side-lobe levels (SLLs) towards the directions of all non-current receiving ABSs, as well as total energy consumption of the AGVs and UAVs, concurrently. Moreover, this problem is proven as a mixed-variable and NP-hard problem.
- *Improved Swarm Intelligence Algorithm:* We propose a swarm intelligence algorithm, namely, improved multi-objective ant lion optimization (IMOALO), to solve the HLMTOP. Specifically, the IMOALO algorithm incorporates chaos solution initialization, dynamic scale factor, and discrete solution update operators to improve the quality of initial solutions as well as balance the exploration and exploitation abilities.
- *Extensive Simulations:* Simulation results show that our proposed DCB-based Approach obviously outperforms other baselines in improving the transmission rate and reducing interference. Moreover, it is found that the proposed algorithm demonstrates strong effectiveness and adaptability across different scales of AGV-UAV networks. In addition, the UAVs and AGVs usually move toward the ABSs and concentrate, thereby obtaining a shorter transmission path and lower SLL.

The remainder of this paper is structured as follows: Section II reviews the related work. Section III presents the system models, and the HLMTOP is formulated in Section IV. Section V shows simulation results, and Section VI provides the discussions. Finally, Section VII concludes this work. Note that this paper is an extended version of [1].

II. RELATED WORK

Recently, there has been a notable emphasis on AGV networks, intelligent transportation, UAV-assisted terrestrial networks, and DCB-based communications. We will briefly introduce several pivotal works to underscore our contributions as follows.

A. AGV Networks

There are several studies tackled the challenges associated with the cooperation and deployment of the AGVs and corresponding networks. Specifically, Liu et al. [10] introduced the multi-AGV real-time collaborative operation method, which utilizes an enhanced A* algorithm to facilitate autonomous and conflict-free navigation for over 100 AGVs. Moreover, Zang et al. [11] developed a trajectory planning optimization

framework to achieve optimal, collision-free, and smooth path planning for multiple AGVs in dynamic and unstructured environments. Furthermore, Li et al. [12] employed an event-based dissipative fuzzy control strategy to improve the path-tracking capabilities of AGVs, thereby underscoring the significance of system stability. In [13], the authors developed a novel mixed integer linear programming model for the integrated scheduling of machines and AGVs in flexible job-shop environments, addressing scalability issues by decomposing the problem into four sub-problems and validating its superior performance through benchmark comparisons and a real-world case study. Elsis et al. [14] introduced a novel IoT paradigm for indoor positioning of AGVs using a deep convolutional neural network (CNN), which converts RSSI signals into 2-D image data, enhancing positioning accuracy and robustness against adversarial attacks. However, none of them consider introducing advanced distributed communicating technology such as DCB to improve the transmission performance.

B. Intelligent Transportation

Many efforts have been made to enhance system efficiency and reliability of intelligent transportation systems. For instance, Zhu et al. [15] integrated dedicated sensing vehicles into sparse mobile crowdsensing systems, devising a hybrid sensing framework to broaden the scope and precision of data collection. Moreover, Kazemi et al. [16] introduced three real-time control strategies predicated on predictive driving conditions, aiming to enhance fuel efficiency. In addition, Qi et al. [17] proposed an integrated connected eco-driving assistance system, focused on achieving improved fuel economy for plug-in hybrid electric vehicles by jointly optimizing vehicle powertrain operations and dynamics. Nevertheless, such systems overlook the utilization of aerial vehicles to assist terrestrial intelligent transportation.

C. UAV-Assisted Terrestrial Networks

Recently, UAVs have played a vital role in assisting terrestrial networks in modern transportation systems. Wang et al. [18] leveraged UAVs as ABSs and introduced the intelligent reflecting surfaces to address signal coverage reliability issues by employing alternating optimization techniques and low-complexity strategies. In [19], the authors introduced a UAV equipped with intelligent reflecting surfaces to tackle communication blockages in densely populated regions, ensuring stable communication links through meticulous two-stage trajectory planning. Moreover, Bansal et al. [20] demonstrated the effective application of rate-splitting multiple access within UAV vehicular communication networks, which reduces interference and augments communication quality. However, none of them uses the close collaborative communication between ground and air equipment to achieve high-speed, low-interference data transmission.

D. DCB Communications

DCB can significantly enhance the communication performance in distributed systems. Specifically, Feng et al. [21]

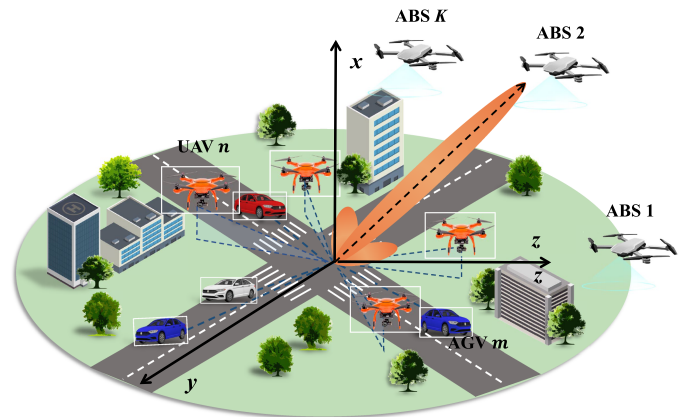


Fig. 1. UAV-assisted AGV data transmission system.

demonstrated that adjusting the flight path of the UAVs and the power settings of ground nodes can improve the data transmission efficiency and mitigate the risk of communication disruptions. Furthermore, Sun et al. [9] utilized an improved multi-objective ant lion optimization algorithm focused on optimizing both time and energy in DCB communications, resulting in a substantial enhancement of energy efficiency. Additionally, Mao et al. [22] developed an innovative approach that combines DCB with distributed backscattering, thus achieving the maximization of overall throughput. Nevertheless, existing works have overlooked the joint deployment of AGVs and UAVs for DCB.

Moreover, some existing works explored the actual deployment of DCB methods. The authors in [23] proposed a DCB framework, SABRE, which synchronizes airborne transmitters for data communication with target receivers. Likewise, the authors in [24] presented an open-source design and implementation, which provides timing, frequency, and phase synchronization for software-defined radios. The authors in [25] represented the first comprehensive algorithmic and systems implementation for air-to-ground DCB on a fleet of UAVs. Such deployment demonstrates the effectiveness of DCB methods in practical applications.

Different from the existing works, we investigate a scenario where the UAVs assist the ground transportation network. Specifically, UAVs aid the ground AGV network in transmitting data to ABSs, and we form an air-ground collaborative VAA by using AGVs and assisting UAVs to facilitate a high-speed and low-interference traffic communication network.

III. SYSTEM MODEL

As illustrated in Fig. 1, we consider a UAV-assisted AGV data transmission system. Specifically, M AGVs, denoted as $\mathcal{A}$, are randomly deployed in a plane area with fixed boundaries, and they can move freely to perform monitoring or sensing tasks. However, the transmit power of a single AGV is limited. In this case, multiple rotary-wing UAVs are dispatched to enhance the efficiency and flexibility of the AGV networks. Note that the UAVs can be divided into two roles according to their functions in this scenario. On the one hand, N UAVs, denoted as $\mathcal{U}$, are used to assist AGVs in performing data transmission. We

TABLE I
SYMBOLS AND DESCRIPTIONS

Symbol	Description
$\mathcal{A}$	The swarm of AGVs.
$\mathcal{U}$	The swarm of UAVs that assists in AGVs data transmission and transmission.
$\mathcal{L}$	The AGV-UAV cluster.
$\mathcal{B}$	The swarm of ABSs.
g_k	The channel power gain.
P_1, P_2	The LoS and NLoS link probabilities.
AF	The array factor of VAA.
$I_{k,w}$	The excitation current weight of the w th AAE for communicating with ABS k .
$p^{k,w}$	The position of the w th AAE for communicating with ABS k .
P_1	The LoS probability of the link.
P_2	The NLoS probability of the link.
R_k	The transmission rate from the VAA to the k th ABS.
E_{AGV}, E_{UAV}	The energy consumption generated by AGV and UAV, respectively.
S	A solution of the HLTMO.
T_{max}	The maximum iteration of the algorithm.
N_{pop}	The population size of the algorithm.
$Archive$	An archive set for storing all the non-dominated solutions.

consider that AGVs and UAVs are both equipped with a single omnidirectional antenna and thus we consider them as a single entity working coordinately, namely, the AGV-UAV cluster. As such, this cluster can be indicated by the set $\mathcal{L} = \mathcal{A} \cup \mathcal{U}$ and $|\mathcal{L}| = W = M + N$. On the other hand, K UAVs, denoted as $\mathcal{B}$, serve as ABSs for receiving and dealing with the data from the AGV-UAV system. Due to the limited transmit power of the system, the AGVs and UAVs within the AGV-UAV cluster will jointly form a VAA and utilize DCB to transmit the collected data to these ABSs. Note that the AGV-UAV VAA needs to communicate with different ABSs for data storage, backup, or edge computing.

Without loss of generality, a three-dimensional (3D) Cartesian coordinate system is considered. Specifically, we use $(x_m^A, y_m^A, 0)$, (x_n^U, y_n^U, z_n^U) and (x_k^B, y_k^B, z_k^B) to denote the positions of the m th AGV, the n th UAV and the k th ABS, respectively. The main symbols used in this work are shown in Table I

A. Joint DCB Model of UAV-AGV Cluster

AGVs and UAVs are typically designed with a focus on lightness, compactness, and high mobility. Therefore, compared to the large antenna systems that are dedicated to communications, the antennas equipped on AGVs or UAVs are usually small. In this case, AGVs and UAVs can be regarded as the distributed antennas of a VAA, and they can closely cooperate to synchronize the timing, initial phase, and frequency, thus achieving a high-gain beam pattern [9]. Different from other many-to-one transmission processes, DCB does not generate additional interference to the destination receiver.

1) *Array Factor of VAA*: The array factor is the sum effect of the radiation field composed by each antenna unit in the antenna array and describes the radiation intensity of the whole antenna system in all directions, which can be represented as

follows [26], [27]:

$$AF(\theta_k, \phi_k) = \sum_{w=1}^W I_{k,w} e^{j \left[\frac{2\pi}{\lambda} (x_{k,w}^L \sin \theta_k \cos \phi_k + y_{k,w}^L \sin \theta_k \sin \phi_k + z_{k,w}^L \cos \theta_k) \right]}, \quad (1)$$

where W is the number of antenna array elements (AAEs). Note that AAE refers to AGV or UAV in this paper. Moreover, $I_{k,w}$ is the excitation current weight, and $p_{k,w}^L$ denote coordinate of the w th AAE for communicating with ABS k , wherein $p_{k,w}^L = (x_{k,w}^L, y_{k,w}^L, z_{k,w}^L)$. In addition, λ indicates the wavelength, and θ_k and ϕ_k denote the elevation and azimuth angles towards ABS k , respectively.

2) *Channel Model*: In some real environments, communication links between the VAA and ABSs may encounter obstacles. To better represent practical scenarios, we adopt a probability line-of-sight (LoS) propagation model. Specifically, we model the LoS and non-line-of-sight (NLoS) propagation based on their occurrence probabilities. The channel power gain is defined as follows:

$$g_k = C_0^{-1} d_k^{-\alpha} [P_1 \zeta_1 + P_2 \zeta_2]^{-1}, \quad (2)$$

where $d_k = \sqrt{(x_o - x_k^B)^2 + (y_o - y_k^B)^2 + (z_o - z_k^B)^2}$ is the distance from the VAA to the k th ABS, wherein (x_o, y_o, z_o) is the coordinate of the VAA center for communicating with ABS k . Moreover, C_0 is a constant related to pathloss, α is the pathloss exponent, and ζ_1 and ζ_2 denote the attenuation coefficients of LoS and NLoS links, respectively. In addition, P_1 and P_2 are the LoS probability and NLoS probability, respectively, i.e., $P_1 + P_2 = 1$. As suggested in [28] and [29], P_1 can be expressed as follows:

$$P_1 = 1 / (1 + C_1 \exp(-C_2[\theta - C_1])), \quad (3)$$

where C_1 and C_2 are the constants associated with the environment. Moreover, θ is the elevation angle between the VAA and the targeted ABS, i.e., $\theta = \frac{180}{\pi} \times \sin^{-1}(\frac{h}{d})$, wherein d and h indicate the horizontal and vertical distance between the VAA and targeted ABS, respectively.

3) *DCB-Based Uplink Transmission*: In the DCB process, all the AGVs and UAVs send the same signal and make the signal superimposed at the target receiver. In this case, all nodes share the same spectrum resources, which means that there is no need to perform bandwidth allocation between different AGV and UAV elements of the VAA in the considered system. As such, based on the array factor model and probabilistic LoS channel model, the transmission rate from the VAA to the k th ABS which is far away from the monitoring area can be given as follows [30]:

$$R_k = B \log_2 \left(1 + \frac{P_t G_k}{g_k \sigma^2} \right), \quad (4)$$

where σ^2 and B are the noise power and transmission bandwidth, respectively, P_t is the total transmit power of the VAA, and G_k is the gain of the VAA towards the location of ABS k , which can be calculated via the array factor by using the method in [30].

B. Energy Consumption Model

Due to the different driving characteristics and energy utilization of AGVs and UAVs, we separately introduce the energy consumption models of AGVs and UAVs, so that providing guidance for designing operation energy optimization strategies.

1) *Energy Consumption Model of AGV*: The sensor, control, and driving systems are the three main components of an AGV. Without loss of generality, we only consider the driving systems of the AGVs since they consume more energy than the sensor and control systems. In this case, the energy consumption of driving systems consists of four key factors, which are kinetic, traction, thermal, and friction, and they can be given by [31]

$$E_{AGV} = E_k + E_e + E_m + E_f, \quad (5)$$

where $E_k = m_{AGV} \times v^2/2$ is the kinetic energy consumption of the AGV, where m_{AGV} is the mass of AGV and v denotes the velocity of the AGV. Moreover, $E_e = \int_0^T (\epsilon \times \tau^2 + \sigma \times v + \lambda \times \tau) d\tau$ is the energy consumption dissipated in the engine in the form of heat, where ϵ and λ are constants associated with the time-heat, and σ is the speed-heat constant. In addition, $E_m = \int_0^T [m_{AGV} \times e^{\zeta\tau} \times \cos(\psi \times \tau + v + (m_{AGV}/2)) + m_{AGV}] d\tau$ is the mechanical dissipation generated by overcoming the friction torque in the actuators, where τ is the stop time of driving. ζ and ψ are the drag and vibration velocity coefficients, respectively. Furthermore, $E_f = \int_0^T (\mu \times m_{AGV} \times v) d\tau$ is the friction consumption during the driving of the AGV, where μ is the friction coefficient between the wheels and the ground.

For AGVs, their motion trajectories are determined based on the position coordinates of the start and end points. Specifically, if the system is just initialized, the AGVs and UAVs may stand by and be randomly distributed in the feasible area, and their positions are the starting positions. As such, when the system has been running for some time, the starting positions represent the positions that the AGVs and UAVs are distributed for performing their previous tasks. Moreover, we consider that the AGVs move at a constant speed on the ground, according to [31]. This is because the AGVs do not require additional energy when they are stationary, and the AGVs have limited flexibility and are not suitable for precise speed control.

2) *Energy Consumption Model of UAV*: The energy consumption of a UAV consists of the propulsion and communication-related energy consumption. In general, the UAVs maintain a hovering state in the air or the flight process will incur a large amount of propulsion energy consumption. According to [32], we ignore the power consumption associated with communication, and the operation energy consumption in the two-dimensional (2D) horizontal plane for a UAV is given by [32]

$$P(v) = \xi_1 (1 + \tau_1 v^2) + \xi_2 \left(\sqrt{1 + \tau_2 v^4} - \tau_3 v^2 \right)^{1/2} + \xi_3 v^3, \quad (6)$$

where v is the flight speed of the UAV. As described in [32], other parameters can be considered as constants. Moreover, the 3D UAV operation energy consumption model can be expressed

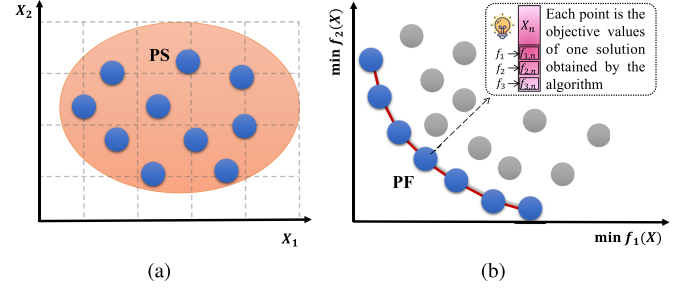


Fig. 2. The example of a two-objective optimization problem.

as follows [29]:

$$E_{UAV} \approx \int_0^T P(v(\tau)) dt + E_1 + E_2, \quad (7)$$

where $v(\tau)$ indicates the UAV velocity at time τ , and T is the end time of a UAV flight. Moreover, $E_1 = \frac{1}{2} m_{UAV} (v(T)^2 - v(0)^2)$ denotes the change of kinetic energy, wherein m_{UAV} represents the aircraft mass. Furthermore, $E_2 = m_{UAV} g (h(T) - h(0))$ denotes the change of potential energy, wherein g represents the gravitational acceleration, and $h(\tau)$ indicates the UAV altitude at time τ .

Unlike other applications, the DCB method requires all UAVs to transmit data from fixed positions simultaneously. This unique requirement means that even if a UAV reaches its designated position early, it must wait until all UAVs are ready to transmit. Thus, for UAVs, we adopt a moving strategy which is similar to [9]. Specifically, the UAVs first fly horizontally and then vertically to their target positions. The speed control scheme follows our previous work in [9], [32]. Note that this scheme is specifically designed for the DCB method, which assigns each UAV an energy-minimizing speed for constructing the VAA according to the time of performing the VAA. By using the speed control scheme, we can ensure that the UAVs fly to the assigned positions in an energy-efficient manner.

C. Multi-Objective Optimization Model

The considered system involves several conflicting objectives. In multi-criteria decision-making, the decision-makers have to consider two or more evaluation metrics or objectives, which may be conflicting or have different importance. In this case, a set of optimal solutions that have an optimal trade-off among multiple objectives can be found by formulating and solving the multi-objective optimization problem (MOP). Mathematically, the MOP with N_{obj} optimization objectives is defined by [33]:

$$\begin{aligned} \min \quad & F = \{f_1(X), f_2(X), \dots, f_{N_{obj}}(X)\}, \\ \text{s.t.} \quad & X \in \Omega, \end{aligned} \quad (8)$$

where $f_i(X)$ indicates the i th optimization function, and $X = \{x_1, x_2, \dots, x_d\}$ denotes the decision variable vector, in which d is the dimension of the decision variables. Additionally, Ω is the set of all the feasible regions. Fig. 2 displays an example of a two-objective optimization problem, and some of the key concepts in MOP are introduced as follows:

Pareto Dominance: Unlike the direct adoption of comparison operators to determine the superiority or inferiority of two solutions in the single objective optimization problem, Pareto dominance is adopted in MOP to prioritize different solutions. The related introduction is as follows:

$$\begin{cases} f_i(X_1) \leq f_i(X_2), & \forall i \in 1, 2, \dots, N_{obj} \\ f_j(X_1) < f_j(X_2), & \exists j \in 1, 2, \dots, N_{obj} \end{cases}, \quad (9)$$

where X_1 and X_2 are any two different solutions in the search space. Specifically, (9) shows that X_1 dominates X_2 (denoted as $X_1 \prec X_2$) if X_1 and X_2 are equal on all objectives and X_1 is superior to X_2 on at least one of the objectives.

Pareto Optimal Solution: If no solution dominates X_1 in the whole search space, X_1 is defined as a non-dominated solution, also called Pareto optimal solution, and all Pareto optimal solutions will form a Pareto set (PS).

Pareto Front: The set of objective functions corresponding to the Pareto optimal solutions is the Pareto front (PF).

IV. PROBLEM FORMULATION

In this section, we delve into an uplink transmission MOP within the UAV-assisted AGV data transmission system to improve energy efficiency while minimizing interference. This necessitates enhancing the directivity of the mainlobe towards the targeted ABS. Carefully optimizing the excitation current weights and positions of AGVs and UAVs is vital for this purpose. However, due to the energy sensitivity of AGV and UAV systems, frequent position changes entail significant energy consumption, thereby shortening the network lifetime. Hence, energy efficiency considerations are integral to driving and flight designs. Additionally, it is important to note that the mainlobe of the VAA can focus only on one targeted receiver at a time, while the data acquisition must be disseminated to multiple ABSs. Therefore, the AGVs and UAVs need to be re-positioned after completing the last data transmission task. Thus, the design of the data transmission order is crucial as it impacts the energy consumption of both AGVs and UAVs [9].

Accordingly, we use $\mathbf{P}^{B \times L} = (\mathbf{X}^{B \times L}, \mathbf{Y}^{B \times L}, \mathbf{Z}^{B \times L})$ and $\mathbf{I}^{B \times L}$ to denote the 3D coordinates and excitation current weights of AAEs for transmitting with the K ABSs, respectively. Moreover, $\mathbf{Q}^{B \times 1}$ indicates the communication order with different ABSs. Specifically, $\mathbf{Q}^{B \times 1} = \{q_1, q_2, \dots, q_K | k \in \mathcal{B}, q_k \in \mathcal{B}\}$. For instance, $\mathbf{Q} = \{6, 3, \dots, 2\}$ indicates the communication with ABS 6 first, followed by ABS 3, and so forth. Thus, $\mathbf{S} = [\mathbf{P}^{B \times L}, \mathbf{I}^{B \times L}, \mathbf{Q}^{B \times 1}]$ is a solution of the HLTMOPT, and it is given by

$$\mathbf{S} = [\mathbf{P}^{B \times L}, \mathbf{I}^{B \times L}, \mathbf{Q}^{B \times 1}],$$

$$= \begin{bmatrix} p_{1,1}^L \cdots p_{1,W}^L & I_{1,1}^L \cdots I_{1,W}^L & q_1 \\ p_{2,1}^L \cdots p_{2,W}^L & I_{2,1}^L \cdots I_{2,W}^L & q_2 \\ \vdots & \vdots & \vdots \\ p_{K,1}^L \cdots p_{K,W}^L & I_{K,1}^L \cdots I_{K,W}^L & q_K \end{bmatrix}, \quad (10)$$

$\underbrace{\hspace{10em}}_{\mathbf{P}^{B \times L}} \quad \underbrace{\hspace{10em}}_{\mathbf{I}^{B \times L}} \quad \underbrace{\hspace{2em}}_{\mathbf{Q}^{B \times 1}}$

Following this, the optimization objectives of the HLTMOPT are presented as follows.

Optimization Objective 1: To improve the communication efficiency of the considered system, we need to minimize the total data transmission time from the VAA to all ABSs, which can be accomplished by jointly optimizing the excitation current weights and positions of AAEs and the communication sequence with various ABSs. Accordingly, the first objective function is given by

$$f_1(\mathbf{P}^{B \times L}, \mathbf{I}^{B \times L}, \mathbf{Q}^{B \times 1}) = \sum_{k=1}^K T_k^{tran}, \quad (11)$$

where $T_k^{tran} = D_k/R_k$ is the transmission time spent from VAA to the k th ABS, in which D_k indicates the total data that the VAA transmits to ABS k .

Optimization Objective 2: We aim to reduce the interference of the AGV-UAV VAA for the non-targeted ABSs. In DCB systems, SLL refers to the intensity of the radiation pattern outside the mainlobe of an antenna array (or VAA). As such, high SLLs can cause significant electromagnetic interference to other communication devices operating in nearby frequencies or spatial regions. By reducing SLL, we can minimize the leakage of signal power into these undesired directions, thereby decreasing interference with other systems. Thus, the second optimization objective is to decrease the intensity of the SLLs in the direction of all non-targeted ABSs, which is given by

$$f_2(\mathbf{P}^{B \times L}, \mathbf{I}^{B \times L}, \mathbf{Q}^{B \times 1}) = \sum_{k=1}^K \sum_{b \in \mathcal{B}-k} \frac{AF(\theta_{SL_b}, \phi_{SL_b})}{AF(\theta_{ML_k}, \phi_{ML_k})}, \quad (12)$$

where $(\theta_{ML_k}, \phi_{ML_k})$ and $(\theta_{SL_b}, \phi_{SL_b})$ refer to the directions of the mainlobe and other sidelobes when the VAA communicates with the current targeted ABS, respectively.

Optimization Objective 3: The AAEs must move to an optimal position to perform the VAA for transmitting data to various ABSs, which will lead to an increase in propulsion energy consumption of the AGV-UAV system. Thus, we need to minimize the total energy consumption of the whole AGV-UAV system to increase the network lifetime, and thus the corresponding objective function is designed by

$$f_3(\mathbf{P}^{B \times L}, \mathbf{Q}^{B \times 1}) = \sum_{k=1}^K \sum_{w=1}^W E_{k,w}, \quad (13)$$

where $E_{k,w}$ is the energy consumption generated by the w th AAE when communicating with the k th ABS.

Accordingly, the HLTMOPT is formulated as follows:

$$\min_{\mathbf{S}} F = \{f_1, f_2, f_3\}, \quad (14a)$$

$$\text{s.t. } C1 : 0 \leq I_{k,w}^L \leq 1, \forall k \in \mathcal{B}, \forall w \in \mathcal{L}, \quad (14b)$$

$$C2 : L_{\min} \leq x_{k,w}^L \leq L_{\max}, \forall k \in \mathcal{B}, \forall w \in \mathcal{L}, \quad (14c)$$

$$C3 : L_{\min} \leq y_{k,w}^L \leq L_{\max}, \forall k \in \mathcal{B}, \forall w \in \mathcal{L}, \quad (14d)$$

$$C4 : H_{\min} \leq z_{k,w}^L \leq H_{\max}, \forall k \in \mathcal{B}, \forall w \in \mathcal{L}, \quad (14e)$$

$$C6 : \mathbf{Q}^{B \times 1} \in \mathcal{Q}, \quad (14f)$$

$$C7 : D_{w_1, w_2} \geq \max\{D_{\min}, \lambda/2\}, \forall w_1, w_2 \in \mathcal{L}, \quad (14g)$$

$$C8: E_{\min} \leq h_k \leq E_{\max}, \forall k \in \mathcal{B}, \quad (14h)$$

where $L_{\max}$ is the maximum range of mobile positions of AAEs in the horizontal direction, and $H_{\min}$ and $H_{\max}$ are the minimum and maximum altitudes that UAVs can fly in the vertical range, respectively. Moreover, $E_{\min}$ and $E_{\max}$ are the minimum and maximum altitudes of ABS in the vertical range, respectively. In addition, due to the mainlobe of VAA can point only in one direction, we use $\mathcal{Q}$ to indicate the set of communication sequences between the VAA and K different ABSs, which has $K!$ possible permutations. Furthermore, the distance between two adjacent moving nodes must be more than $D_{\min}$, which is a constraint condition defined in (14g). Likewise, for the cases of highly dense AGVs and UAVs in a small area, the effect of mutual coupling cannot be ignored, especially when the distance between two neighboring antennas is less than $\lambda/2$. This is because mutual coupling between the AGVs and UAVs in the VAA may cause the beam pattern to be significantly damaged compared to that of the AGVs and UAVs taken in isolation. Thus, constraint (14g) also ensures that the minimum separation distance is greater than $\lambda/2$. As can be seen, the problem has mixed-type and large-scale decision variables, and also the following properties.

Lemma 1: The formulated HLTMOOP shown in (14) is NP-hard.

Proof: We first simplify HLTMOOP by assuming that the AGVs and UAVs have fixed excitation current weights and positions [9]. In this case, f_3 can be simplified as follows:

$$\begin{aligned} \min_{S(\mathcal{Q}^{B \times 1})} f &= \sum_{k=1}^{K-1} E_{q_k, q_{k+1}}, \\ \text{s.t. } \mathcal{Q}^{B \times 1} &\in \mathcal{Q}, \end{aligned} \quad (15)$$

where $E_{q_k, q_{k+1}}$ is the operation energy consumption caused by AGVs and UAVs during the transition from communication with the q_k th ABS to the q_{k+1} th ABS. The simplified f_3 shown in (15) is a traveling salesman problem (TSP) [34], [35], which has been proved to be NP-hard. Thus, the formulated HLTMOOP shown in (14) is an NP-hard problem. ■

As can be seen, the formulated optimization problem is an NP-hard MOP with conflicting optimization objectives. In this case, one feasible method is to transfer the multiple objectives of the MOP to a single-objective optimization problem by combining different optimization objectives (e.g., using the weighting method) and then using a single-objective optimization algorithm to solve it. However, the weights among different optimization objectives are usually difficult to determine [36]. In the following, we aim to introduce multi-objective optimization algorithms to solve the MOP directly, thereby flexibly obtaining the appropriate solution regardless of scenario changes.

V. ALGORITHM DESIGN

HLTMOOP has mixed-type and large-scale decision variables, and it is also a complex NP-hard problem, which is difficult to solve by using traditional optimization methods such as convex optimization. In this case, evolutionary computation,

with its stochastic nature inspired by natural selection, which is a potent tool for tackling NP-hard problems. Specifically, these algorithms have the ability to explore diverse solution spaces, coupled with scalability and adaptability, making them well-suited for navigating the complexities of HLTMOOP. By harnessing evolutionary algorithms, we can efficiently search for globally optimal or near-optimal solutions even in highly nonlinear and multimodal structures of HLTMOOP.

Among various evolutionary computation algorithms, MOALO [37] has several advantages in tackling optimization problems. First, it leverages the collective intelligence of ant lion colonies, mimicking their foraging behavior to efficiently explore solution spaces and converge to Pareto optimal solutions. Second, MOALO incorporates multiple objectives jointly, allowing it to generate diverse, well-distributed solutions along the Pareto front, providing decision-makers with a comprehensive view of trade-offs. Finally, MOALO offers robustness and adaptability, adjusting its search strategy dynamically to handle various problem landscapes effectively, making it suitable for complex real-world optimization tasks. The details of MOALO are introduced as follows.

A. Conventional MOALO

In conventional MOALO, the position of each ant represents a feasible solution, and the global search of the solution space is achieved by simulating the random wandering behavior of ants. Moreover, the traps created by the antlion can affect the random wandering routes of the ants, in which the antlion refers to the local optimal solution and it can be replaced by the ant with a better objective function value. Fig. 3 gives the framework of MOALO, and the main steps are corresponding details.

Step 1: MOALO utilizes a random strategy to initialize the positions of multiple ants, i.e., a set of feasible solutions to MOP, denoted as Pop .

Step 2: Calculate the corresponding function values of each solution in Pop on the three optimization objectives according to (11)–(13).

Step 3: Prioritize solutions in Pop based on the objective values calculated above and the concept of Pareto dominance introduced in Section III-C.

Step 4: The solutions in Pop need to be updated to find more excellent solutions in the global search space. Accordingly, the position update method of an ant is as follows:

$$S_l^t = \frac{R_A^t + R_E^t}{2}, \quad (16)$$

where S_l^t denotes the position of the l th ant at the t th iteration. Moreover, R_A^t and R_E^t indicate the random walk around the antlion selected by the roulette wheel and the random walk around the elite antlion (the antlion with the best objective function value) in t th iteration, respectively.

Step 5: Determine whether to continue the iteration based on the algorithm termination conditions, such as whether the maximum number of iterations has been reached.

Despite the superior performance of the MOALO algorithm, it still faces several significant challenges in dealing with the

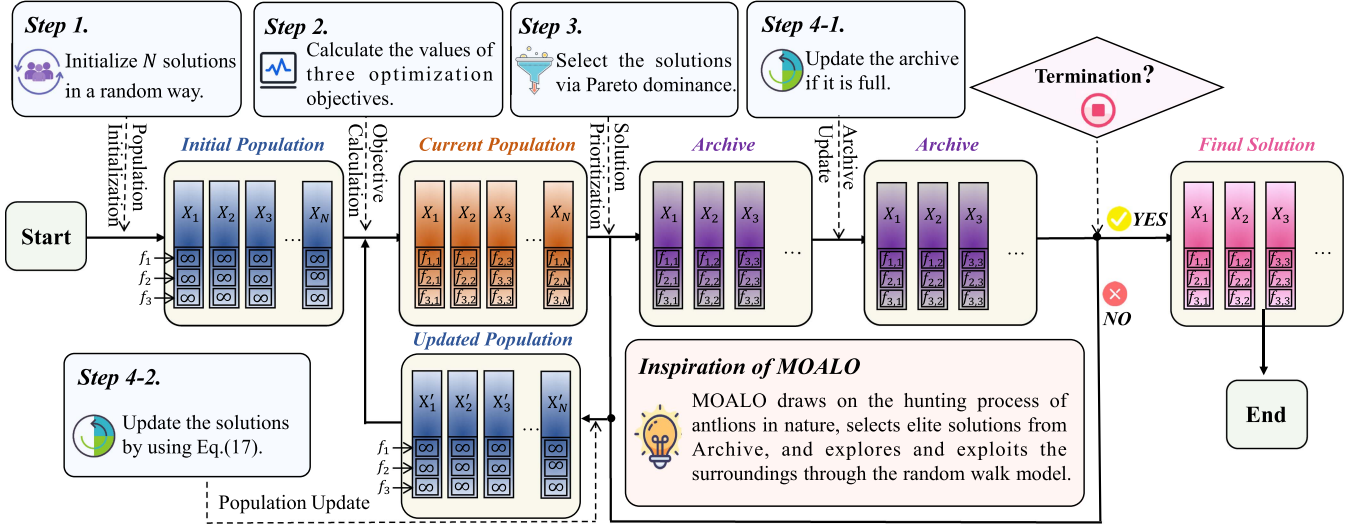


Fig. 3. The main framework of MOALO.

formulated HLTMO. For example, HLTMO is a hybrid optimization problem with large-scale decision variables. Thus, conventional MOALO is unable to optimize the discrete solution space directly, and it is difficult to determine all decision variables simultaneously. Moreover, MOALO may fail to converge or fall into a local extremum during the process of solving HLTMO. Therefore, we need to improve conventional MOALO to obtain better solutions.

B. Imoalo

To address the aforementioned challenges, we propose an IMOALO with three enhanced operators, i.e., chaos solution initialization, dynamic scale factor, and discrete solution update strategies. The pseudocode of the proposed IMOALO is presented in Algorithm 1.

1) *Chaos Solution Initialization*: The quality of the initial solutions affects the optimization capability of conventional MOALO. In this case, random initialization may result in a low-performance distribution of the initial population, thus affecting the convergence performance of MOALO. Moreover, some low-quality solutions in the initial population may increase the search space complexity and reduce the search efficiency of MOALO. Thus, it is essential to choose a suitable strategy to improve the initial solution distribution. Among various optimization strategies, chaotic mapping has the characteristics of nonlinearity, ergodicity, and randomness, which can obtain a set of initial solutions with an even distribution by mapping the candidate solutions in the initial solution space to the chaotic space. Therefore, we adopt the Logistic map to optimize the initial solutions, which can be expressed as follows [38]:

$$h_{i+1} = ah_i(1 - h_i), \quad (17)$$

where h_i is the i th value of the chaotic array generated by (17), and $a \in [0, 4]$ is the control parameter. Moreover, Fig. 4 shows the visual model of the Logistic map when the maximum number of iterations is set to 200.

Algorithm 1: IMOALO.

Input: maximum iteration T_{max} , population size N_{pop} and archive set $Archive$, etc.;

Output: updated archive set $Archive$;

```

1 for  $l = 1$  to  $N_{pop}$  do
2   Initialize  $S_l$  by using Eq. (18); //Solution
   initialization operator based on Logistic map
3 end
4 for  $t = 1$  to  $T_{max}$  do
5   for  $l = 1$  to  $N_{pop}$  do
6     Select randomly a solution from the  $Archive$ ;
       //  $R_A^t$ 
7     Select a solution from the  $Archive$  according
       to the roulette wheel strategy; //  $R_E^t$ 
8     Update  $(P^{B \times L}, I^{B \times L})$  of the solution  $l$  by
       using Eq. (19); //Update the continuous
       decision variables of a solution
9     Update  $(Q^{B \times 1})$  of the solution  $l$  according to
       Algorithm 2; //Update the discrete decision
       variables of a solution
10  end
11  Calculate the objective function values of all
     solutions and update the  $Archive$ ;
12  if  $Archive$  is full then
13    Delete some solutions from the  $Archive$ 
       according to the roulette wheel strategy;
14  end
15  Return  $Archive$ ;
16 end

```

Accordingly, the chaos solution initialization operator can be described as follows:

$$S(j) = LB_j + h_j \times (UB_j - LB_j), \quad (18)$$

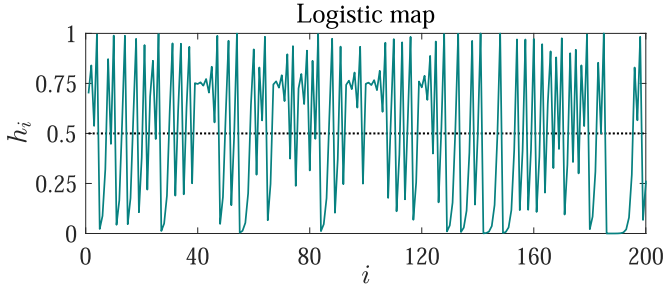


Fig. 4. Logistic map visualization. As can be seen, the Logistic map mixes a certain degree of randomness and periodicity. This characteristic is conducive to helping the algorithm find a high-performance initial solution in the solution space [39].

where $S(j)$ is the j th dimension of a solution, and UB_j and LB_j indicate the upper and lower bounds of the j th dimension, respectively.

2) *Dynamic Scale Factor*: In evolutionary computation, the exploration capability is the ability of the algorithm to leave the current extremes and search for a globally optimal solution, while the exploitation capability describes the ability of the algorithm to approach a better solution by gradually optimizing based on the optimal solution already found. Thus, adjusting the ratio between the exploration and exploitation phases is the key to improve the optimization performance of an evolutionary algorithm.

According to (16), the update of ant positions is guided by the antlion R_A , which is selected according to the roulette wheel selection strategy and elite antlion R_E . Moreover, the scale factors of R_A and R_E are the same and remain constant during the whole iteration process, which may result in a decrease in the optimization capability of conventional MOALO. Therefore, we introduce a dynamic scale factor to balance the exploration and exploitation capabilities of MOALO, which can be described as follows:

$$S_l^t = \mu_1 R_A^t + \mu_2 R_E^t, \quad (19)$$

where μ_1 and μ_2 are the dynamic scale factors of R_A and R_E , respectively, which can be calculated as follows:

$$\mu_1 = 0.5 - 0.05 * (1 - e^{-\frac{2.5t}{T}}), \quad (20)$$

$$\mu_2 = 0.5 + 0.05 * (1 - e^{-\frac{2.5t}{T}}), \quad (21)$$

As can be seen, μ_1 and μ_2 change dynamically with iteration, which makes the two random walks occupy different weights at different periods during the operation of IMOALO. Specifically, IMOALO gradually increases the influence of the elite antlion on positional updates during the optimization process, while decreasing the weighting of the random walk style of the antlion selected by the roulette wheel. As such, we can ensure the global search ability in the early phase of IMOALO and the convergence accuracy in the late phase of IMOALO.

3) *Discrete Solution Update*: The conventional MOALO can only deal with the continuous optimization problems, while the formulated HLTMO is a mixed-type optimization problem with a continuous solution space ($\mathbf{P}^{B \times L}$, $\mathbf{I}^{B \times L}$) and a discrete

Algorithm 2: Discrete Solution Update Strategy.

Input: S_l
Output: S_l'
1 **if** $rand > 0.7$ **then**
2 | Update ($Q^{B \times 1}$) of S_l by using PMX operator;
3 **end**
4 **else if** $rand < 0.3$ **then**
5 | Update ($Q^{B \times 1}$) of S_l by using EM operator;
6 **end**
7 **Return** S_l' ;

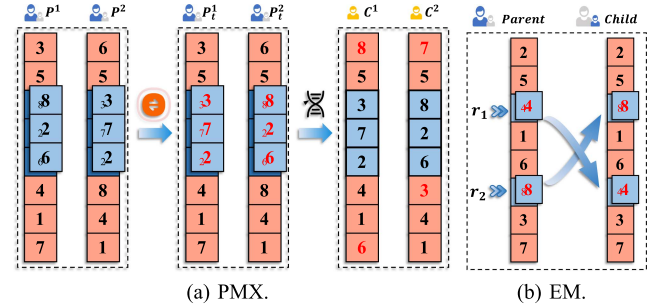


Fig. 5. Sketch map of the PMX and EM.

solution space ($Q^{B \times 1}$). Thus, we propose a discrete solution update operator to overcome this issue.

To update the discrete component of a solution, we introduce the partially mapped crossover (PMX) [40] and exchange mutation (EM) [41] strategies, and the details are as follows:

PMX: The sketch map of the PMX is shown in Fig. 5(a), and the PMX operator is briefly presented as follows:

First, sub-fragments of the same size are selected at the same positions of the two discrete solutions (such as the blue part of P^1 and P^2 in Fig. 5(a)). **Second**, the selected sub-fragments are exchanged while the other parts remain fixed, resulting in two temporary discrete solutions P_t^1 and P_t^2 . **Third**, a mapping table needs to be generated for conflict detection based on the selected sub-fragments (For example, 8 with 3, 7 with 6 in Fig. 5(a)). **Finally**, the unexchanged parts of P_t^1 and P_t^2 are changed according to the mapping table until each temporary discrete solution contains no duplicate values, thus obtaining the corresponding child individuals C^1 and C^2 .

EM: As shown in Fig. 5(b), two positions r_1 and r_2 of the solution *parent* are randomly selected. Then, the values at these two positions are exchanged to achieve mutation of discrete dimensions.

Accordingly, Algorithm 2 presents the main steps of the discrete solution update operator.

4) *Complexity Analysis*: The main computation of the IMOALO occurs in the iteration phase. In each iteration, population Pop and archive set $Archive$ need to be updated and maintained. Let the number of optimization objectives be N_{obj} , the size of Pop be N_{pop} , and the capacity of the $Archive$ be N_{pop} , then the computational cost of updating the Pop is $O(N_{pop})$. Moreover, the process of maintaining the $Archive$ will involve the computation of the objective functions and the

TABLE II
SIMULATION PARAMETER SETUPS

Parameters	Description	Values
M	The number of AGVs	4, 8
N	The number of UAVs	4, 8
K	The number of ABSs	8
α	Pathloss exponent	2
P_{AAE}	Transmit power of each AGV or UAV	0.1 W
D_{min}	Collision distance	0.5 m
H_{min}	The minimum flight altitude of UAV	75 m
H_{max}	The maximum flight altitude of UAV	95 m

TABLE III
ALGORITHM PARAMETER SETUPS

No.	Algorithms	Parameters
1	MODA [43]	$w \in [0.4, 0.9]$, $s \in [0, 0.2]$, $a \in [0, 0.2]$, $c \in [0, 0.2]$, $f \in [0, 0.2]$, $e \in [0.01]$
2	MOMVO [44]	$p = 6$
3	MOPSO [45]	$w = 0.5$, $c_1 = 1$, $c_2 = 2$
4	MOALO [42]	$w \in [2, 3, 4, 5, 6]$, $\zeta_1 = 0.5$, $\zeta_2 = 0.5$
5	IMOALO	$w \in [2, 3, 4, 5, 6]$, $\zeta_1 \in [0.5, 0.45]$, $\zeta_2 \in [0.5, 0.55]$

ordering of the non-dominated solutions, in which the computational cost of the objective functions is $\mathcal{O}(N_{obj} \times N_{pop})$ and the complexity produced by the non-dominated sorting process is $\mathcal{O}(N_{obj} \times N_{pop} \times \log N_{pop})$. Therefore, the complexity of the IMOALO algorithm is $\mathcal{O}(N_{obj} \times N_{pop}^2)$.

VI. SIMULATION RESULTS AND ANALYSIS

In this section, we conduct simulations to evaluate the performance of the proposed IMOALO for dealing with the formulated HLTMOF.

A. Simulation Setups

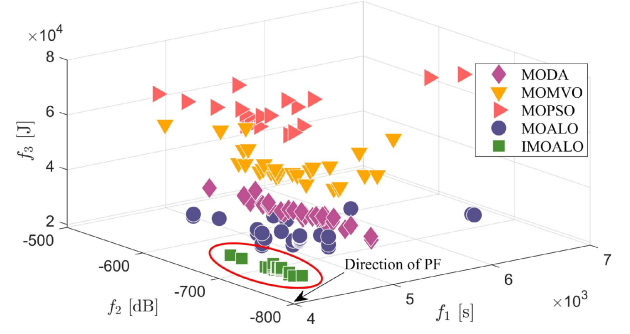
In the simulations, the monitor area A is set to be $100 \text{ m} \times 100 \text{ m}$ [27]. Additionally, the key parameters in the DCB-based data transmission network are shown in Table II. The parameters of the AGVs and UAVs of this work follow [31] and [32], respectively. Moreover, we select the starting positions of AGVs and UAVs by randomly generating positions from the feasible motion area of the AGV and UAV, respectively.

Moreover, we introduce the conventional MOALO [42], multi-objective dragonfly algorithm (MODA) [43], multi-objective multi-verse optimizer (MOMVO) [44], and multi-objective particle swarm optimization (MOPSO) [45] as the comparison algorithms. The maximum number of iterator T and the population size N_{pop} of these algorithms are set as 500 and 30, respectively. Moreover, other key parameter settings are given in Table III.

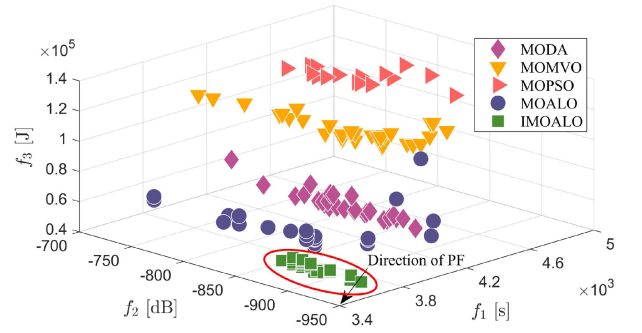
B. Results and Analysis

To verify the performance of the proposed algorithm, two data transmission networks with 8 AAEs and 16 AAEs are considered.

Fig. 6(a) and (b) show the Pareto solution distributions obtained by the IMOALO and other comparison approaches for dealing with the formulated HLTMOF in the cases of 8 AAEs



(a) 8-AAE case.



(b) 16-AAE case.

Fig. 6. Solution distributions obtained by various algorithms.

and 16 AAEs, respectively. It can be seen from the figures that the solutions achieved by the IMOALO are closer to the direction of the PF, indicating that the proposed IMOALO outperforms other algorithms in solving HLTMOF. Moreover, there are no extreme solutions in the PS obtained by the proposed IMOALO compared with other algorithms, demonstrating the excellent convergence accuracy of the proposed IMOALO algorithm.

Fig. 7(a) and (b) show the optimization results achieved by various algorithms regarding the total transmission time (f_1), total SLLs in the direction of non-current targeted ABSs (f_2), and total energy consumption of AGVs and UAVs (f_3) in different cases. Note that the SLLs are typically measured in decibels (dB), which is a logarithmic scale used to represent the ratio of the power of a signal compared to a reference level. Thus, the SLLs are negative values that indicate suppression of sidelobe power relative to the mainlobe. As can be seen, the proposed IMOALO algorithm outperforms other comparison algorithms on three optimization objectives in the two cases. The improved performance can be attributed to the chaos solution initialization operator, which enhances the diversity of the initial solutions, thus improving the global search performance of the algorithm. Moreover, the dynamic scale factor strategy effectively balances the global exploration and local exploitation abilities of the algorithm, further enhancing the corresponding solving performance.

Fig. 8 shows the moving path of AAEs obtained by the proposed IMOALO algorithm for transmitting the collected data to the first ABS in the 8-AAE and 16-AAE cases. As can be

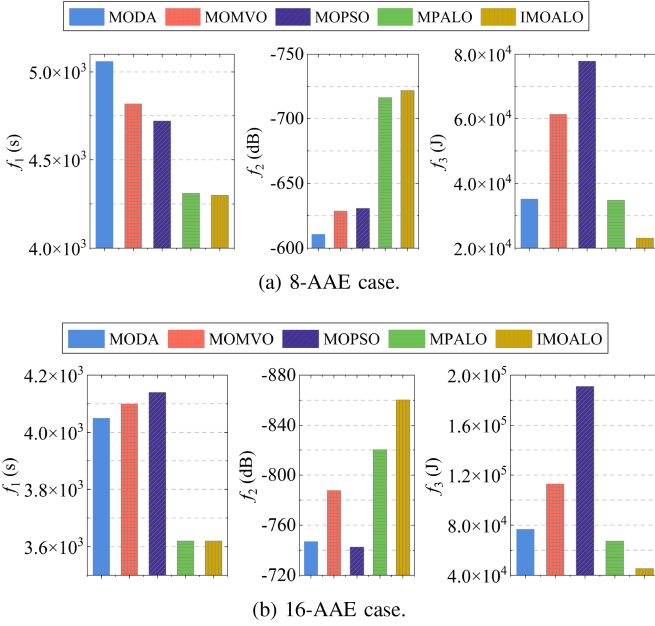


Fig. 7. The optimization results achieved by various algorithms regarding the total transmission time (f_1), total SLLs in the direction of non-current targeted ABSs (f_2), and total energy consumption of AGVs and UAVs (f_3).

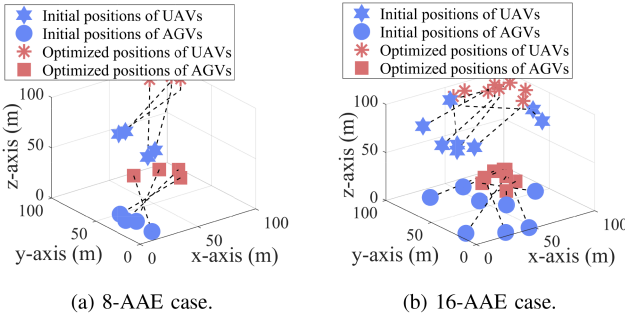


Fig. 8. Moving path of the AAEs obtained by the IMOALO algorithm for communicating with the first ABS.

seen, the short moving path of AGVs and UAVs reflects the well-optimized energy efficiency of the AGV-UAV system.

VII. DISCUSSION

We present the discussion on the applicability and usability of the proposed method for practical applications.

A. Impact of Synchronization Failures

To address potential failures or errors in the synchronization algorithm, the following measures can be used to reduce the impacts:

- **Redundancy Mechanisms:** To mitigate the impact of synchronization failures, the system is designed with redundant UAVs or AGVs. These additional UAVs and AGVs can assume control in the event of a failure of the primary unit. This ensures that the system maintains coordinated operation even if an error occurs in the synchronization mechanism of a specific UAV or AGV.

- **Error Detection and Recovery:** Real-time error detection mechanisms are integrated at the receiver to continuously monitor the synchronization status. If any discrepancies are detected, the receiver triggers an error and sends a message to the AGV-UAV system for re-synchronization. This process involves protocols that recalibrate the timing and position data of the AGVs and UAVs, ensuring alignment with the mission objectives.

B. Precise Control of Multiple UAVs and AGVs

To achieve precise control of multiple UAVs, we present two different methods, each suitable for specific scenarios.

- **Scenario 1: Algorithm Running on UAV or AGV:** If the proposed algorithm is executed on one of the UAVs or AGVs, the results of the algorithm can be distributed to other UAVs and AGVs through a self-organizing UAV-AGV network. Upon receiving the results, the other UAVs or AGVs can use existing control methods and modules to move to the designated positions, thereby ensuring coordinated movement and alignment among all units based on the computed results.
- **Scenario 2: Algorithm Running on Ground Station:** If the proposed algorithm is executed on a ground station, the results are broadcast to all UAVs and AGVs via a wireless network. Upon receiving the broadcast results, the UAVs and AGVs can then utilize their existing control methods and modules to move to their designated positions, thus ensuring that all UAVs and AGVs follow the ground station's instructions accurately and maintain the desired formation or trajectory.

C. Mechanism of Selecting Final Solution

In the considered multi-objective optimization, the process typically generates a set of Pareto-optimal solutions, each representing a different trade-off among the objectives. The selection of the final solution from this set depends on the specific requirements and priorities of the practical application.

Selection Criteria: The final solution is chosen based on the preferences of the decision-makers, who may prioritize certain objectives over others. For instance, in applications where energy efficiency is critical, a solution that minimizes energy consumption might be preferred, even if it results in a slightly longer transmission time or higher sidelobe levels.

The Role of Decision-Makers: In practical applications, the final decision is usually made by stakeholders or experts who understand the operational context and objectives of the system. They evaluate the trade-offs presented by the Pareto-optimal solutions and select the one that best aligns with their goals and constraints. They can also establish rules to automatically select the final decision.

D. Impacts of Some Special Situations

In actual operation, UAVs may confront issues such as swarm heterogeneity and deviation from the original position, which are discussed as follows.

On the one hand, the swarm may contain heterogeneous UAVs. However, the proposed method remains applicable as long as the communication equipment operates within the same frequency band. Specifically, the proposed method is primarily based on DCB and the 3D mobility of UAVs. This approach requires that all UAVs and AGVs function as elements within a virtual antenna array. Thus, if the system operates under these conditions, the proposed method can still be effective. Therefore, the appearance, energy, and power of the UAVs will not impact the applicability of our method, provided that the communication devices are equipped with omnidirectional single antennas and operate within the same communication frequency band.

On the other hand, UAVs may experience slight displacement from their original positions due to factors such as wind, which could potentially cause minor disruptions in the beamforming pattern. However, due to the precise positioning and adjustment capabilities of modern UAVs, any displacement caused by wind is typically minimal, and its impact on the system's performance is not significant. This conclusion is supported by simulation in the literature, such as references [8], [9], [46]. Moreover, when the wind subsides, UAVs can quickly return to their designated positions due to their high maneuverability. Therefore, even if a UAV is temporarily displaced by wind, the performance of the proposed method is not significantly affected.

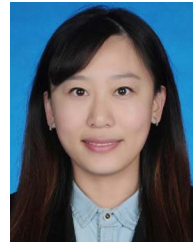
VIII. CONCLUSION

In this study, we proposed a novel approach employing a swarm of UAVs to augment terrestrial AGV networks. We formulated an HLTMOPT aimed at minimizing the uplink transmission time, the total SLLs towards non-current receiving ABSs, and the operation energy consumption of the AGVs and UAVs. To address the NP-hard nature of HLTMOPT, we proposed an IMOALO algorithm with three enhancements that are the chaos solution initialization, dynamic scale factor adjustment, and discrete solution update mechanism. Simulation results demonstrated that the proposed IMOALO outperforms other evolutionary computation methods, including MODA, MOMVO, MOPSO, and conventional MOALO, across all optimization objectives, showcasing its effectiveness in solving complex optimization challenges in AGV-UAV networks. Moreover, the UAVs and AGVs usually move toward the ABSs and concentrate, thereby obtaining a shorter transmission path and lower SLL. In the future, we will focus on optimizing UAV flight trajectories by discretizing time to tackle prevalent real-time operation challenges within AGV-UAV networks.

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